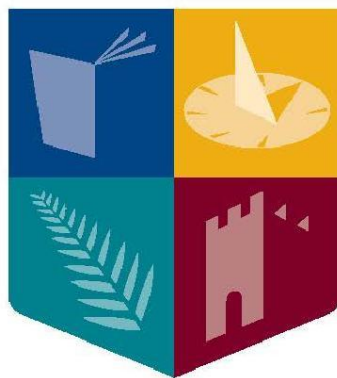


EE204 Amplifiers
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Table of Contents

32.2(a) Common emitter Amplifier.....	2
Introduction	2
Theory	2
Circuit Description	2
Results	3
Conclusion.....	4
Problem 32.2(b)	5
Introduction	5
Theory	5
Circuit Description	5
Results	6
Conclusion.....	7

32.2(a) Common emitter Amplifier

Introduction

The aim of Problem 32.2(a) is to identify the amplifier configuration of the given circuit and to examine its response to a small sinusoidal input signal. A time-domain simulation is used to observe the input, base, and output waveforms and to verify the expected behaviour of the amplifier. In particular, the presence of voltage gain and phase inversion is investigated.

Theory

The circuit shown in Problem 32. 2(a) corresponds to a common-emitter amplifier. In this configuration, the emitter terminal is common to both the input and output reference, the input signal is applied to the base, and the output is taken from the collector

When the transistor is biased in the active region, a small change in base–emitter voltage produces a proportional change in collector current. This varying collector current flows through the collector resistor, producing a voltage variation at the collector. As the collector voltage decreases when the collector current increases, the output signal is inverted by 180° relative to the input.

Input and output coupling capacitors are used to block DC while allowing AC signals to pass. The base bias resistor establishes a steady DC operating point so that the transistor remains in the active region throughout the signal cycle, allowing linear small-signal amplification.

Circuit Description

The circuit was implemented and simulated in LtSpice as showcased in Figure 1 – Circuit implementation of problem 32.2(a) on LtSpice . A 10V DC (V2) voltage was used to bias the transistor. A sinusoidal signal of 10mv amplitude and 5khz frequency was applied through the base coupling capacitor (C1).

Voltages were then measured at three key point. Where each point was displayed on a separate plot pane to clearly observe the small input and larger output.

- Input (Left of C1)
- Base node (between C1, R1 & T)
- Collector node (between Q1, R2 & C2)

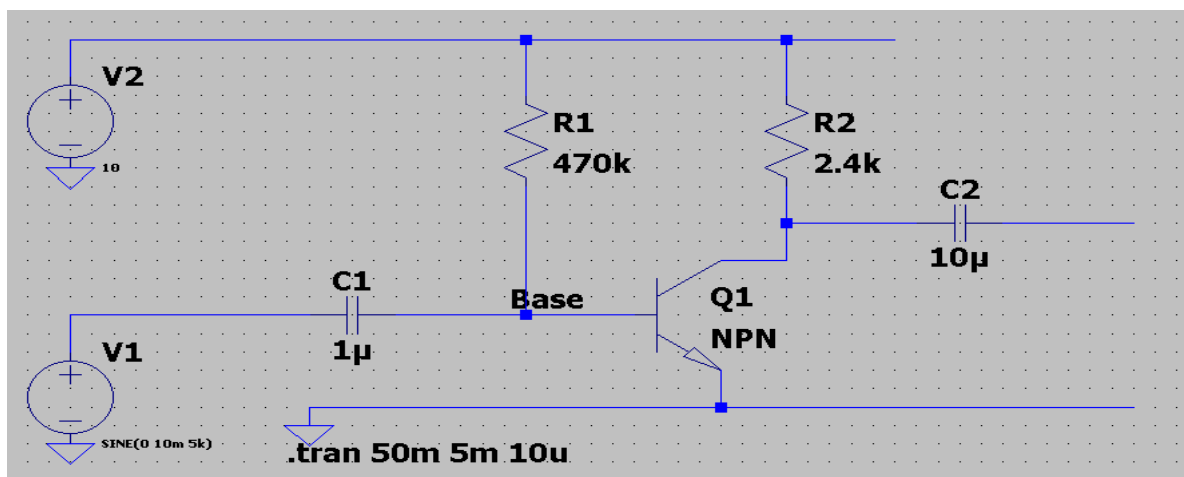


Figure 1 – Circuit implementation of problem 32.2(a) on LtSpice

Results

The input and base waveforms obtained from the simulation in LtSpice are shown in Figure 2 – input and base waveforms. The input signal is a sinusoidal waveform with an amplitude of approximately ± 10 mV and frequency of 5 kHz centred at 0V. The base voltage is approximately 0.78 V with a small sinusoidal variation, showing that the transistor is correctly biased and responds to the input signal.

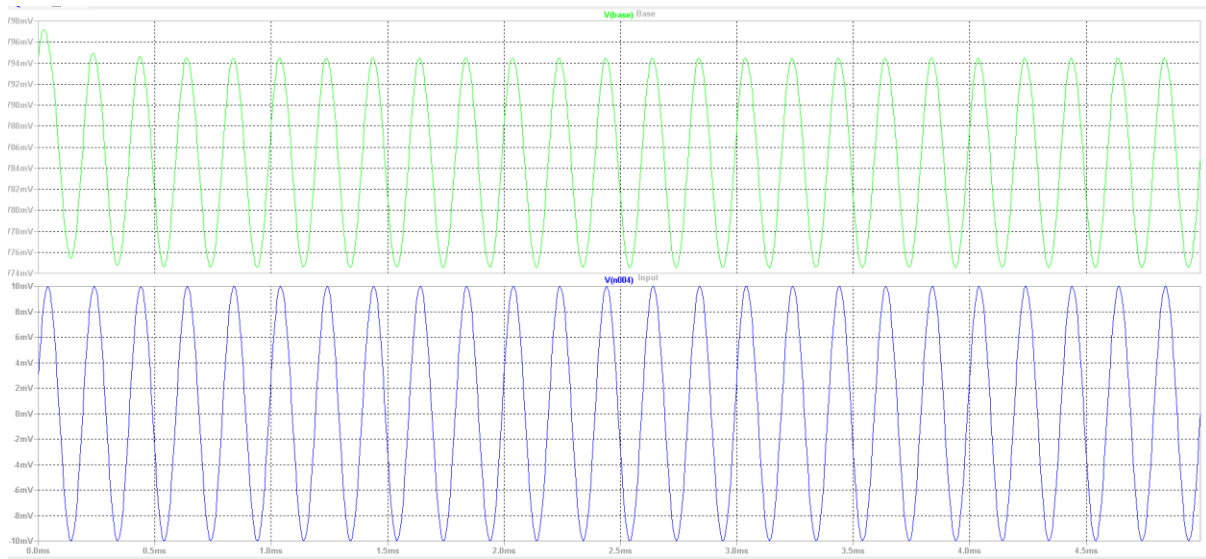


Figure 2 – input and base waveforms (a)

The collector waveform obtained from the simulation in LtSpice is shown in Figure 3 – collector node waveform. The output shows a much larger amplitude compared to the input signal and it is also inverted 180° relative to the input signal. Therefore, the collector voltage demonstrates the expected voltage gain and phase inversion of a common emitter amplifier.

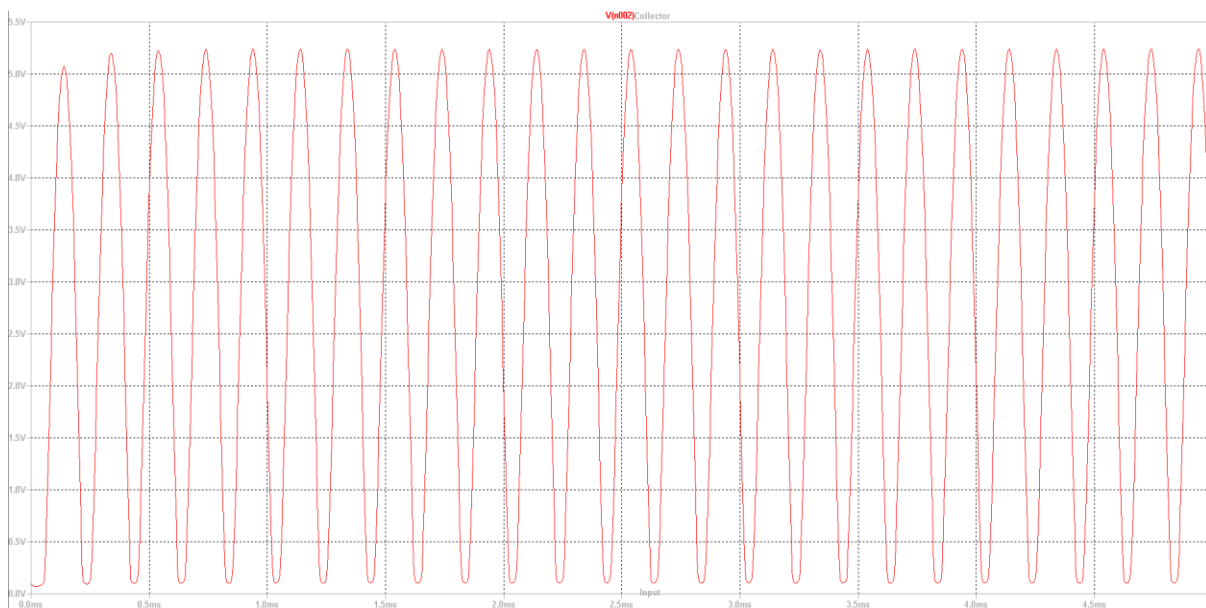


Figure 3 – collector node waveform (a)

Conclusion

In this experiment a common emitter transistor amplifier was analysed and simulated using LtSpice. A small sinusoidal input was successfully amplified and inverted at the collector. As when simulated they closely matched theoretical expectations confirming the fundamental characteristics of a common emitter amplifier.

Problem 32.2(b)

Introduction

The aim of problem 32.2(b) is to analyse a modified transistor amplifier circuit and examine its response to the input signal. Compared to part(a) this circuit uses collector to base feedback biasing to improve bias stability where it was simulated on LtSpice to confirm the amplifier operation.

Theory

The circuit in problem 32.2(b) is also a common emitter amplifier. As in part (a) the emitter is connected to ground and the input signal is applied to the base through the capacitor and the output is taken from the collector.

In this configuration a resistor is connected to the collector and base creating negative feedback for DC biasing. If the collector increases the collector voltage drops, reducing the base voltage through the feedback resistor which stabilises the operation point and makes the circuit less sensitive to transistor parameter variations.

Although this circuit uses a different biasing method it uses the same concept as the previous circuit. Variations in the base voltage create changes in the collector current which converts it to an output voltage across the collector resistor. As a result, the output signal amplifies and inverted with proportion to the input.

Circuit Description

This circuit was implemented in LtSpice using a 9V DC supply. A sinusoidal input signal of 10mV amplitude and 5kHz frequency was applied to the base through the coupling capacitor. The collector to base resistor provided biasing without the use of a separate base voltage divider.

Voltages were then measured at three key point. Where each point was displayed on a separate plot pane to clearly observe the small input and larger output.

- Input (Left of C1)
- Base node (between C1, R1 & T)
- Collector node (between Q1, R2 & C2)

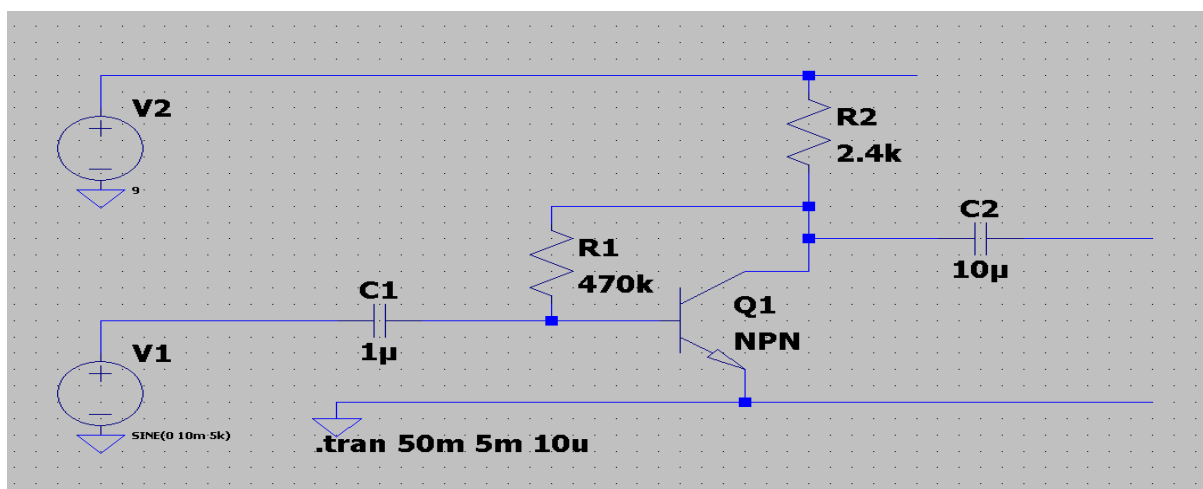


Figure 4 – Circuit Implementation of problem 32.2(b) on LtSpice

Results

The input waveform was observed as a small sinusoid of approximately ± 10 mV, centred at 0 V. The collector waveform, shown in Figure 4 – Circuit Implementation of problem 32.2(b) on LtSpice, showcased a significantly larger amplitude and was inverted by 180° relative to the input signal. The collector voltage remained biased at an intermediate DC level, indicating that the transistor was operating in the active region without saturation or cutoff.

These results confirm that the circuit in Problem 32.2(b) operates as a common-emitter amplifier, with collector-to-base feedback providing stable biasing while preserving voltage gain and phase inversion.

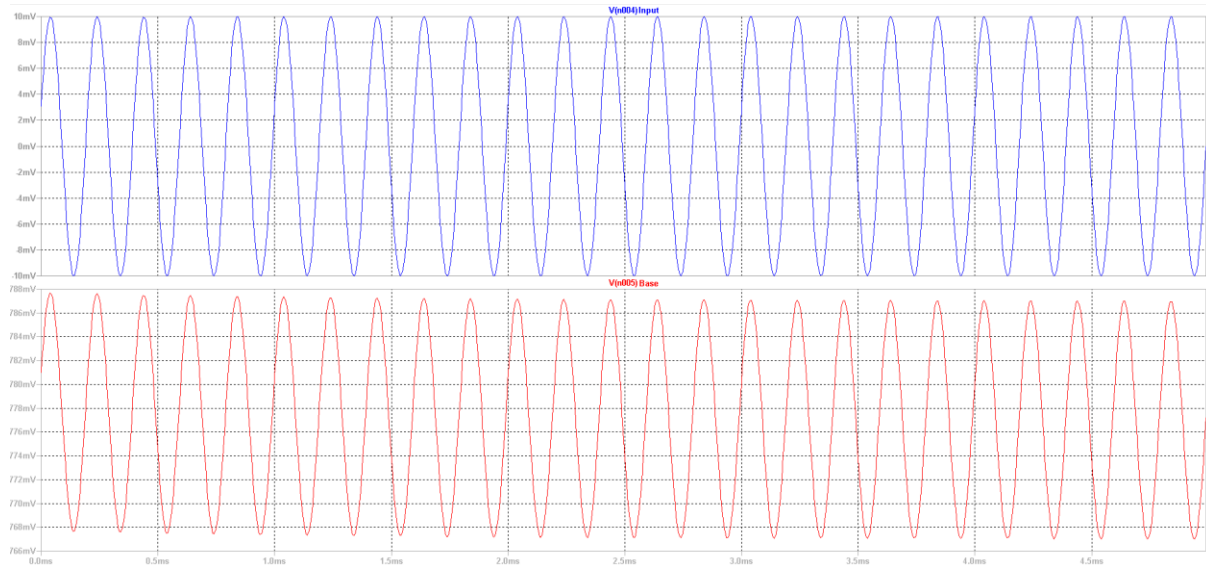


Figure 5 – input and base waveforms (b)

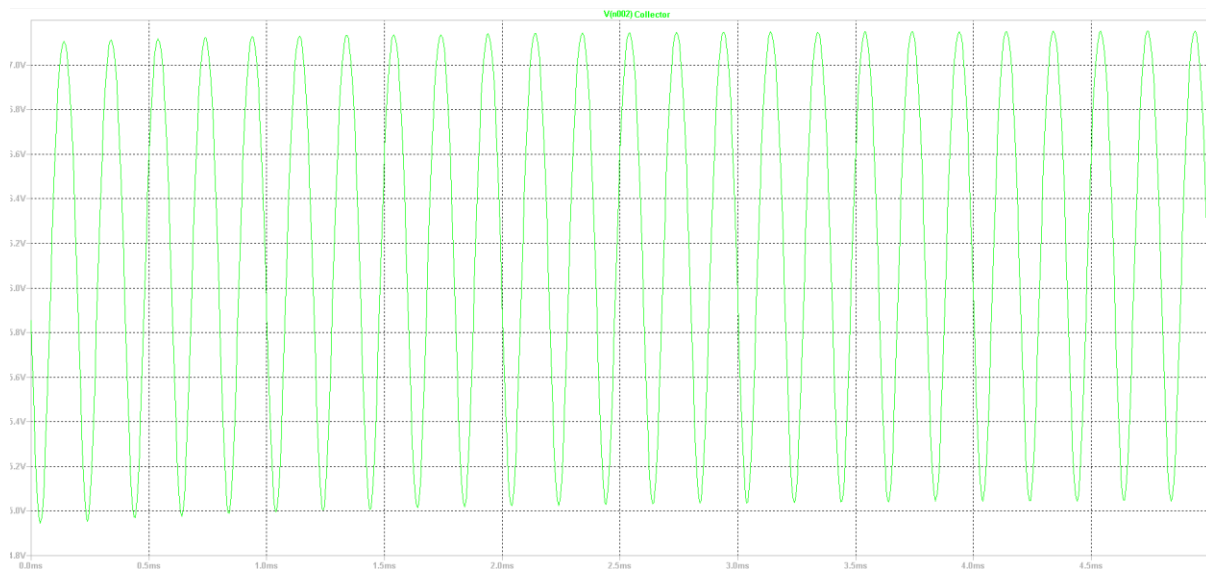


Figure 6 – Collector waveform (b)

Conclusion

In this experiment, a common-emitter amplifier using collector-to-base feedback biasing was analysed and simulated. Compared to the circuit in Part (a), this biasing method provides improved stability of the operating point while maintaining the same fundamental amplifier behaviour. The simulation results showed that the output at the collector was amplified and inverted by 180° relative to the input signal, confirming correct common-emitter operation. Overall, the results are consistent with those obtained in Part (a), while demonstrating the advantage of feedback biasing in improving bias stability.